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Nutrition Society Journal Club (NSJC)

Food Systems Theme

***Increasing production diversity and diet quality:
Evidence from Bangladesh***



COI's

The speaker declares no conflicts of interest

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Increasing production diversity and diet quality: Evidence from Bangladesh

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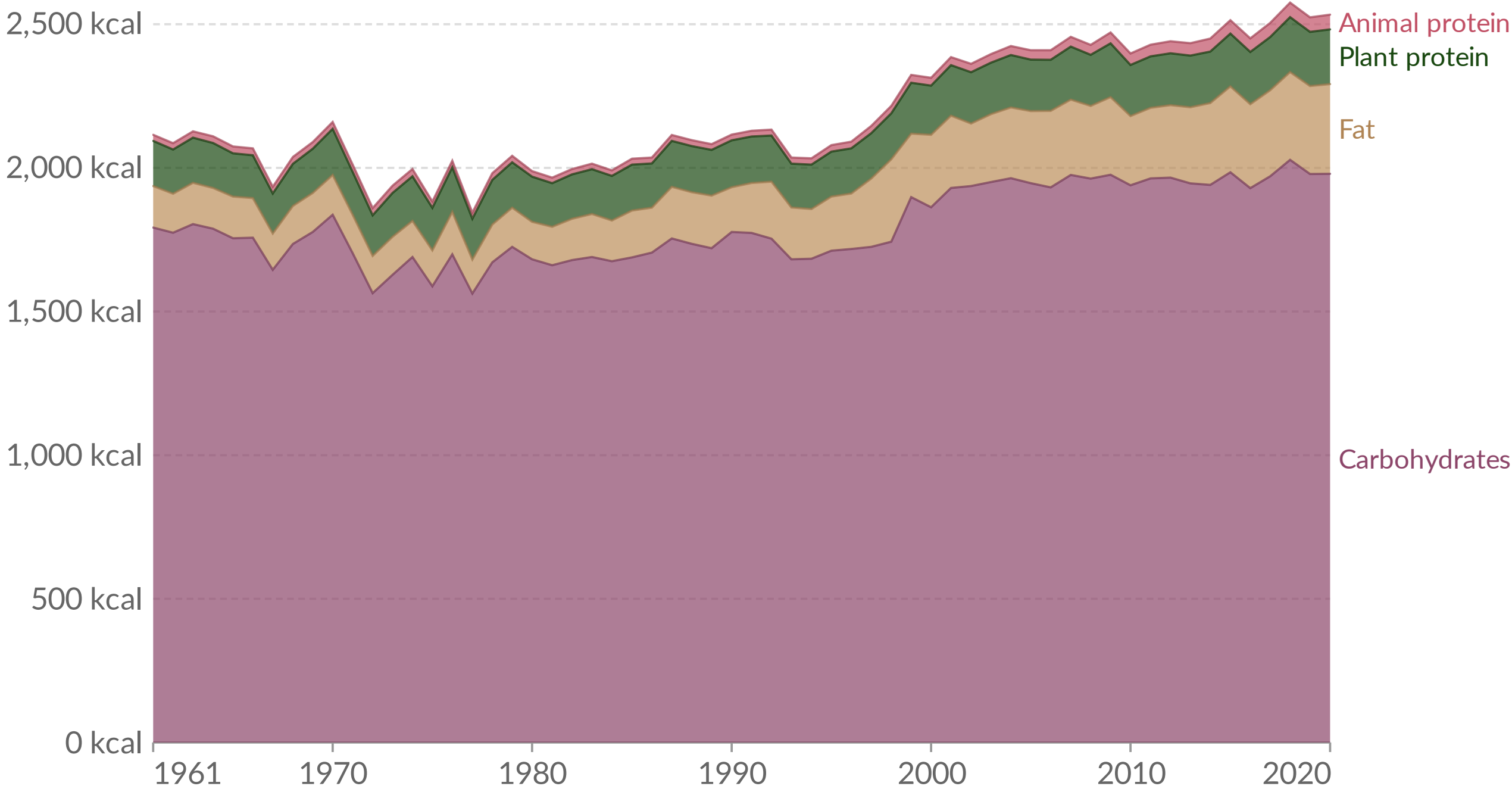
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Prompt for observations

Daily caloric supply derived from carbohydrates, protein and fat, Bangladesh, 1961 to 2020



The average per capita supply of calories derived from carbohydrates, protein and fat, all measured in kilocalories per person per day.



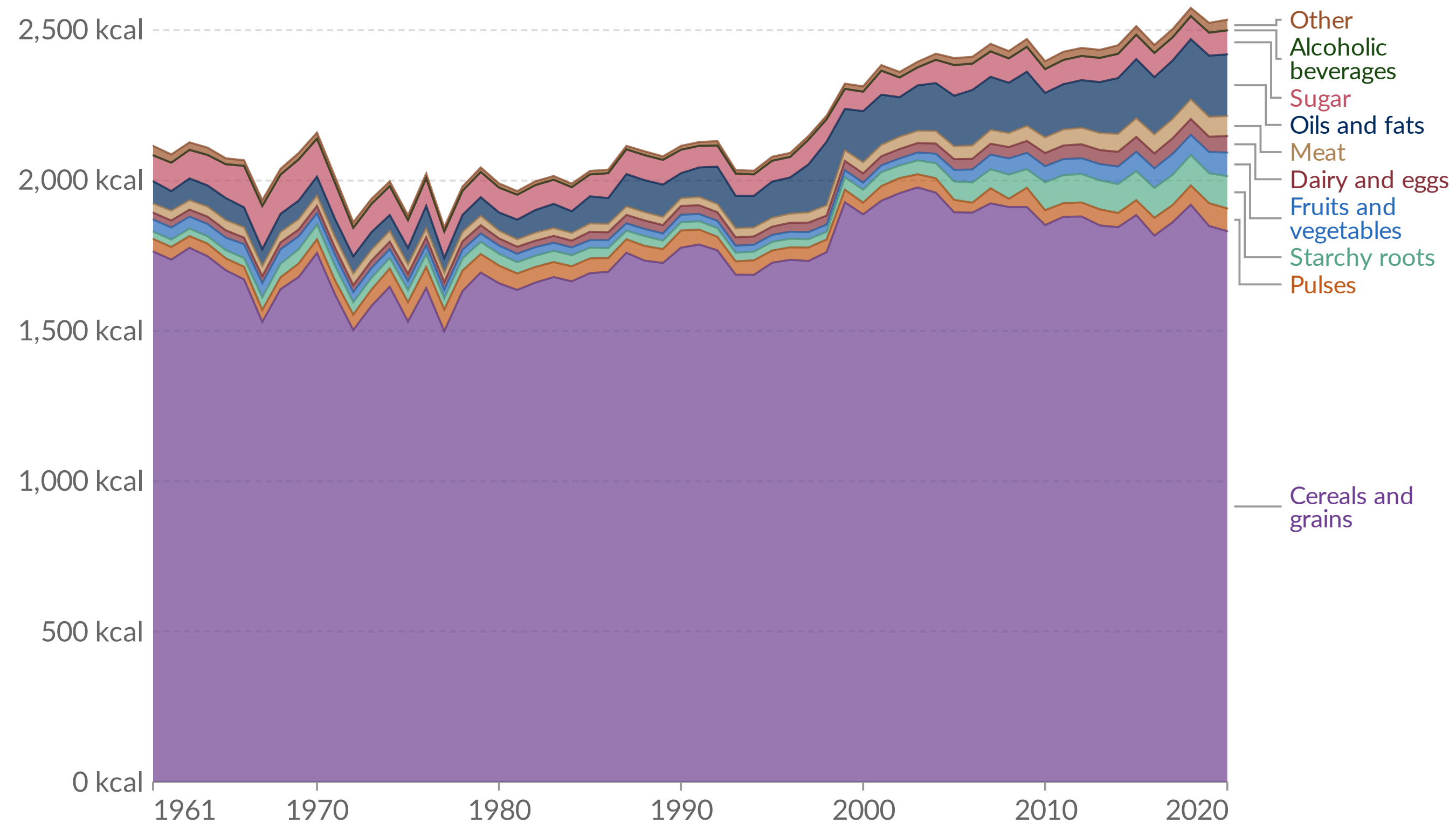
Data source: Our World in Data based on the Food and Agriculture Organization of the United Nations
OurWorldInData.org/diet-compositions | CC BY



Dietary compositions by commodity group, Bangladesh, 1961 to 2020

Our World
in Data

Average per capita dietary energy supply by commodity groups, measured in kilocalories per person per day.



Data source: Food and Agriculture Organization of the United Nations

OurWorldInData.org/diet-compositions | CC BY



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Diet quality matters in Bangladesh and beyond

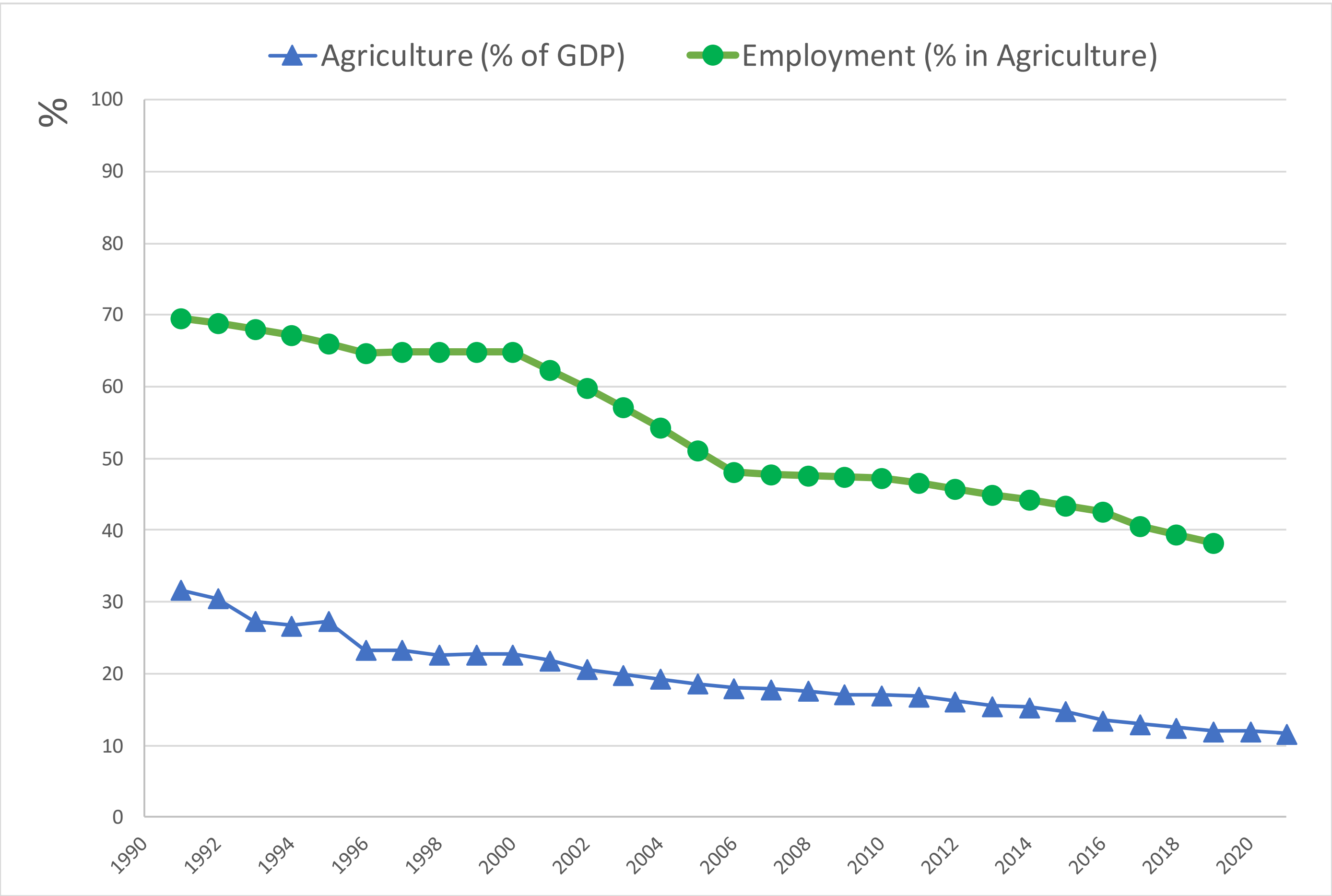
- Diets are limited in fruits, vegetables, pulses, and animal-sourced foods
- Perishability and food safety is a challenge
- Poor quality diets cause malnutrition
 - Poor growth in children
 - Micronutrient deficiencies
 - Unwanted weight gain

Diversifying production may improve dietary quality

- Dietary diversity
- Women's empowerment



Farming and farmers in Bangladesh, 1990 – 2021



Magnitudes are unknown, especially within particular contexts, and other factors play a role

- How strong is the relationship between production diversity and dietary diversity?
- What other interventions can help leverage the relationship between production diversity and dietary diversity?
- Can intervention(s) be scaled-up?



The Bangladesh MoA implemented the Agriculture, Gender, and Nutrition Linkages (ANGeL) project

- Cluster randomized control **(C)** trial design
- Treatment arms:
 - Nutrition behavior change communication **(T-N)**
 - Agriculture training **(T-A)**
 - Agriculture training + nutrition behavior change communication **(T-AN)**
 - Agriculture training + nutrition behavior change communication + gender sensitization **(T-ANG)**
- Questions
 - Do any of the treatment arms improve production diversity?
 - Do any of the treatment arms improve dietary diversity / quality?
 - Do combined arms have larger impact?

The ANGeL project had several features that were unique

- Implemented by the Bangladesh Ministry of Agriculture
- Husbands and wives were trained together
- Project implemented across 100 different subdistricts in rural Bangladesh
- Robust measurement
 - Fields and homesteads
 - New agricultural land planted? Or increasing output within a given area?
 - Dietary diversity, quantity of food, and dietary quality
 - Global Diet Quality Score
 - Women's Empowerment in Agriculture Index (WEAI) score

What did each of the treatment arms look like?

- Nutrition behavior change communication (**T-N**)
 - 19 sessions
- Agriculture training (**T-A**)
 - 17 sessions conducted by MoA Agriculture Officers (agricultural extension agents)
 - High-value crops, crop calendars, small plots/gardens, water, pest, and fertilizer management, livestock, aquaculture, identification of nutrient rich crops
- Agriculture training and nutrition behavior change communication (**T-AN**)
 - 19 sessions from **T-N** + 17 sessions from **T-A** = 36 sessions.
- Agriculture training + nutrition behavior change communication + gender sensitization (**T-ANG**)
 - 36 sessions from T-AN + 8 sessions on gender = 44 sessions
 - Activities that improved respect, appreciation, and communication for families

All 8 administrative divisions of Bangladesh were represented in 16 upazilas (10 blocks per upazila*)

T-A	T-N	T-AN	T-ANG	Control
• 25 blocks • 1 village from each block • 25 households in each village • 625 households	• 25 blocks • 1 village from each block • 25 households in each village • 625 households	• 25 blocks • 1 village from each block • 25 households in each village • 625 households	• 25 blocks • 1 village from each block • 25 households in each village • 625 households	• 35 blocks • 1 village from each block • 25 households in each village • 875 households

Total sample size:

3375 households with 2500 across the 4 treatment arms

* 25 blocks with only gender sensitization activities conducted by community women were not included in the analysis for practical reasons and research-question related reasons

Mechanisms for impact

- Did treatment arms increase knowledge?
 - 20 questions about nutrition knowledge
- Did participants act on this increased knowledge?
 - Did participants increase the consumption of foods produced in home gardens?
 - Did their dietary quality improve?
 - HDDS
 - Consumption quantity
 - Global Diet Quality Score

Household production, farming knowledge, and agricultural practices were measured, as well as household diet and nutrition knowledge

- *Nutrition knowledge*
 - % correct on an assessment about micronutrient-rich foods and food safety and cooking practices
- *Household dietary diversity score (HDDS)*
 - From 7 day recall of household food consumption for: (1) cereals, (2) roots and tubers, (3) vegetables, (4) fruit, (5) meat, poultry, offal, (6) eggs, (7) fish and seafood, (8) pulses, legumes, and nuts, (9) dairy, (10) oil/fats, (11) sugar/honey, (12) other. Indicator ranges from 0 – 12.
- *Per capita caloric acquisition*
 - Quantities from 7-day recall are converted to kcal per capita per day
- *Household Global Diet Quality Score (hGDQS)*
 - 25 food groups (16 “healthy” and 7 “unhealthy” and 2 “unhealthy in excess only”); includes amounts as low / medium / high / very high. Indicator ranges from 0 – 49.
- *Diet of adequate quality*
 - Binary. = 1 if hGDQS is ≥ 23 , =0 otherwise.



Global Diet Quality Score in JNutr (2021)



The Journal of Nutrition

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Development and Validation of a Novel Food-Based Global Diet Quality Score (GDQS)

Bromage Sabri¹  , Batis Carolina², Bhupathiraju Shilpa N^{1 3}, Fawzi Wafaie W¹, Fung Teresa T^{1 4}, Li Yanping¹, Deitchler Megan⁵, Angulo Erick², Birk Nick¹, Castellanos-Gutiérrez Analí², He Yuna⁶, Fang Yuehui⁶, Matsuzaki Mika⁷, Zhang Yiwen¹, Moursi Mourad⁵, Kronsteiner-Gicevic Selma^{1 8}, Holmes Michelle D^{1 3}, Isanaka Sheila¹, Kinra Sanjay⁹, Sachs Sonia E¹⁰...Willett Walter C^{1 3}

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Baseline covariates included

- Age and sex of household head
- Mean educational level for men and women
- Number of adults in the household
- Dependency ratio
- Wealth index
- Access to electricity, cell phones, TV
- Amount of land owned
- Any fishponds owned
- Recent exposure to Agricultural Officers
- Impacted by flooding yes / no



Building rafts for transport in an area on the Brahmaputra River in northern Bangladesh. Two weeks of flooding left half the country under water. KM Asad/NurPhoto, via Getty Images

Flooding in Bangladesh in 2017

Analysis was intent-to-treat (ITT)

- Participants are included in the analysis in the group to which they were originally assigned
- Leakage and mix-ups may have led to people participating in a different arm
 - Not likely an issue for this study due to village-level implementation
- Loss to follow up may be different depending on original treatment group
 - Possibly an issue for this study
- Different levels of adherence depending on original treatment group
 - Possibly an issue for this study
- Using ITT, randomization still valid
- Using ITT, generalizability is better

Treatment arms that included nutrition behaviour change communication increased nutrition knowledge

Treatment arm impact differed

	Nutrition knowledge, % correct
	Women
Treatments	
T-N	2.749*** (0.648)
T-A	0.625 (0.656)
T-AN	3.303*** (0.520)
T-ANG	3.055*** (0.579)

<i>p</i> values, equality of treatments	
T-N = T-A	<0.01
T-N = T-AN	0.32
T-N = T-ANG	0.61
T-A = T-AN	<0.01
T-A = T-ANG	<0.01
T-AN = T-ANG	0.60
Mean, control group	80.1
Observations	3278
R-squared	0.188

All treatments improved HDDS, hGDQS, binary GDQS. Calories increased w/ all arms except T-A.

		Dietary Diversity Score	Log per capita caloric acquisition	Log household Global Diet Quality Score	Diet of adequate quality
Treatments					
T-N	Improved all measures of diet quality but did not affect agricultural production from homesteads (those results not shown here)	0.207**	0.032*	0.065***	0.144***
		(0.079)	(0.017)	(0.011)	(0.030)
			[0.032]	[0.067]	
T-A		0.207***	0.007	0.036***	0.066***
		(0.078)	(0.016)	(0.009)	(0.024)
			[0.007]	[0.037]	
T-AN	Double the impact on HDDS compared with the other arms	0.398***	0.040**	0.070***	0.157***
		(0.073)	(0.016)	(0.011)	(0.027)
			[0.041]	[0.073]	
T-ANG		0.346***	0.059***	0.081***	0.198***
		(0.078)	(0.018)	(0.012)	(0.029)
			[0.061]	[0.084]	
				Largest impact on hGDQS – likelihood of adequate diet increased by 19.8%	

T-A vs. T-AN do not have the same impact

T-AN vs. T-ANG have about the same impact

	Dietary Diversity Score	Log per capita caloric acquisition	Log household Global Diet Quality Score	Diet of adequate quality
<i>p</i> values, equality of treatments				
T-N = T-A	0.99	0.20	0.01	0.01
T-N = T-AN	0.02	0.69	0.73	0.68
T-N = T-ANG	0.10	0.22	0.23	0.10
T-A = T-AN	0.01	0.09	<0.01	<0.01
T-A = T-ANG	0.11	0.01	<0.01	<0.01
T-AN = T-ANG	0.52	0.32	0.46	0.23
Mean, control group (levels)	7.7	1982	22.2	0.42
Observations	3285	3286	3286	3286
R-squared	0.236	0.087	0.260	0.188

Many strengths of the study already discussed, and also include the following

- High attendance at training sessions
 - Median woman attended 79% – 94% of sessions
 - Median man attended 75% – 94% of sessions
 - Participants perceived sessions as accessible and informative
- Only 2.5% of baseline were lost to follow up
 - Migration (N=64)
 - Declined / dropped out / could not find (N=10)
 - Incomplete interview (N=12)

Potential limitations for understanding nutritional status

- Only small correlations between treatment arm and attrition
 - Households in T-AN compared to C were 1.5 percentage points more likely to attrit
 - But cannot reject the null that attrition does not differ across treatment arms
- Attrition is slightly higher in flood-affected (2017) areas, and w/ high dependency ratio
 - Attrition is not associated with other baseline characteristics
- Magnitude of impacts?
 - ~0.30 more food groups
 - Increased hGDQS by ~8% with T-ANG
- Household level dietary quality variables

Production diversification can help improve rural dietary quality for farmers

- Participants were satisfied with the training(s)
- Participants had increased knowledge
- Improved agricultural knowledge led to better agricultural practices
- Impacts were larger for women than men
- Field crops and homestead garden crops should both be considered
 - Small impacts on the extensive margin for agricultural production (field crops)
 - Large effects on the intensive margin for agricultural production (homesteads)
 - Fruit and vegetable production increased 33%
 - Egg production increased 58%
 - Dairy production increased 42%
- Baseline production of eggs and dairy was low
- Bundled treatments did not differ in affecting agricultural production

Households consumed the additional homestead production

- Treatment arms did not differ
- All treatment arms improved diet quality except that T-A did not improve calories
- Including gender sensitization did not have different impacts than other arms
 - Possibly because women and men were trained together? Or fewer sessions?
- Scale-up seems feasible based on large-scale implementation by MoA
- Could the results be seen with fewer sessions?
- What are the positive externalities of the training or of changed practices in the long-run?
- Good potential and impact but not a cure-all

Questions for discussion – please pop your questions and comments into the question box!

- What are the benefits of randomization?
- What are the benefits of including a control group?

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